

COS20019 - Cloud Computing Architecture

Week 4: ACF Lab 4: Working with EBS

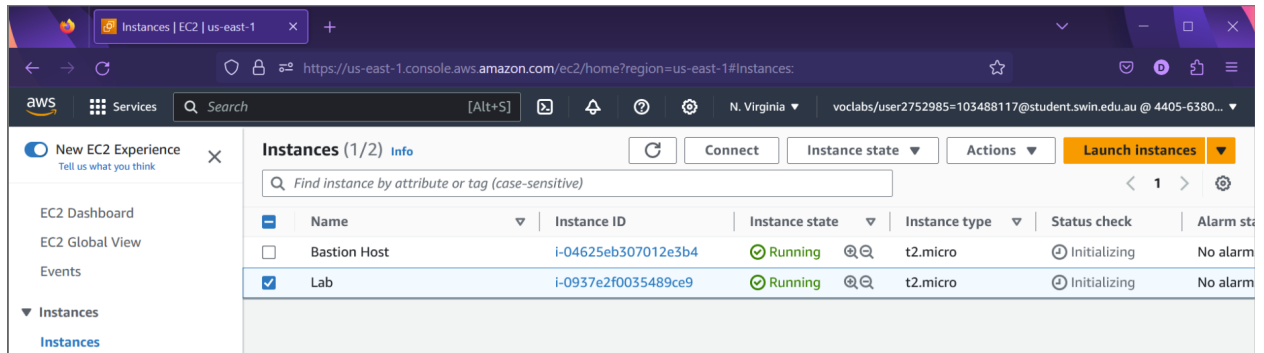
Author: Trac Duc Anh Luong - ID: 103488117

Due Date: 01/10/2023

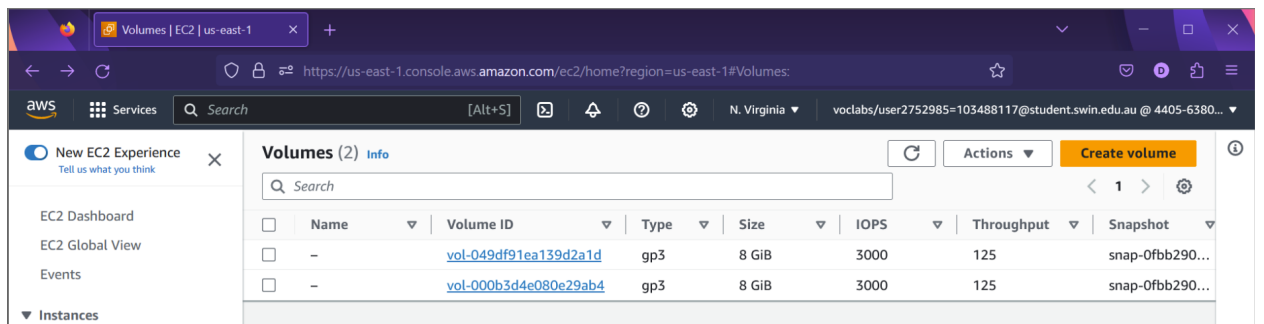
Task 1: Create a New EBS Volume

In this task, you will create and attach an Amazon EBS volume to a new Amazon EC2 instance.

5. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
6. In the left navigation pane, choose **Instances**.
An Amazon EC2 instance named **Lab** has already been launched for your lab.



7. Note the **Availability Zone** of the instance. It will look similar to *us-east-1a*.
8. In the left navigation pane, choose **Volumes**.
You will see an existing volume that is being used by the Amazon EC2 instance. This volume has a size of 8 GiB, which makes it easy to distinguish from the volume you will create next, which will be 1 GiB in size.



9. Choose **Create volume** then configure:
 - **Volume Type:** *General Purpose SSD (gp2)*
 - **Size (GiB):** **1**. **NOTE:** You may be restricted from creating large volumes.
 - **Availability Zone:** Select the same availability zone as your EC2 instance.
 - Choose **Add Tag**
 - In the **Tag Editor**, enter:
 - **Key:** **Name**
 - **Value:** **My Volume**

Create volume | EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateVolume:

aws Services Search [Alt+S]

N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 4405-6380...

EC2 > Volumes > Create volume

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp2)

Size (GiB) [Info](#)

1

Min: 1 GiB, Max: 16384 GiB. The value must be an integer.

IOPS [Info](#)

100 / 3000

Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) [Info](#)

Not applicable

Availability Zone [Info](#)

us-east-1a

Snapshot ID - optional [Info](#)

Don't create volume from a snapshot

Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q My Volume X

Remove

Add tag

You can add 49 more tags.

Cancel

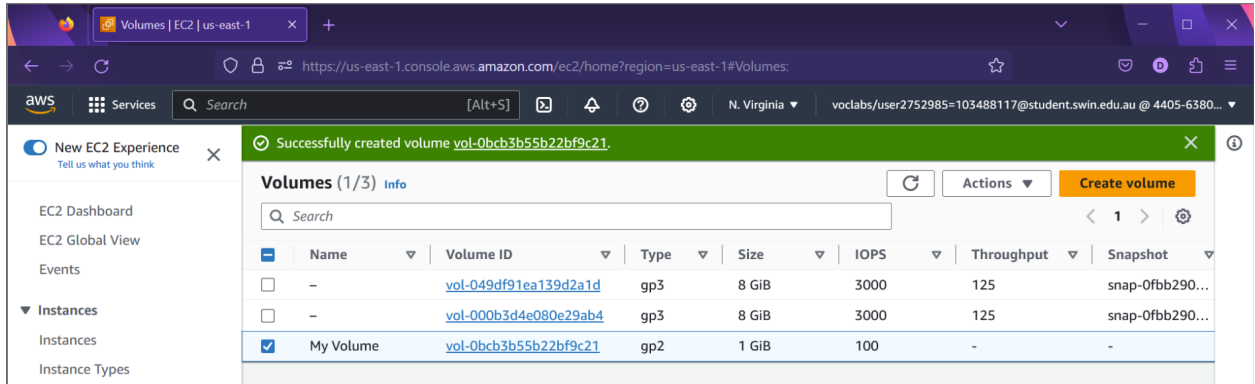
Create volume

CloudShell Feedback

© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

10. Choose Create Volume

Your new volume will appear in the list, and will move from the *Creating* state to the *Available* state. You may need to choose **refresh** to see your new volume.



Successfully created volume [vol-0bcb3b55b22bf9c21](#).

Volumes (1/3) [Info](#) [Refresh](#) [Actions](#) [Create volume](#)

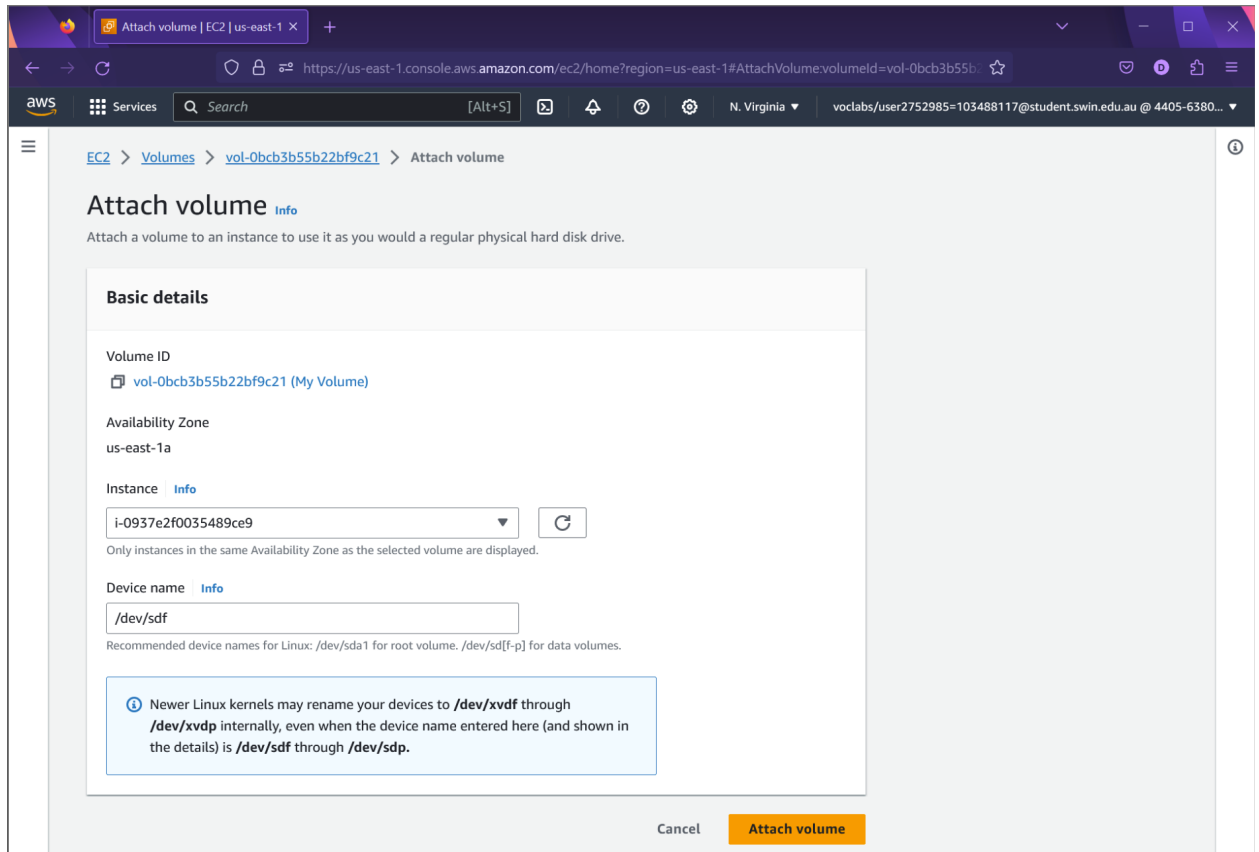
	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot
☐	-	vol-049df91ea139d2a1d	gp3	8 GiB	3000	125	snap-0fbb290...
☐	-	vol-000b3d4e080e29ab4	gp3	8 GiB	3000	125	snap-0fbb290...
☒	My Volume	vol-0bcb3b55b22bf9c21	gp2	1 GiB	100	-	-

Task 2: Attach the Volume to an Instance

You can now attach your new volume to the Amazon EC2 instance.

11. Select **My Volume**.
12. In the **Actions** menu, choose **Attach volume**.
13. Choose the **Instance** field, then select the instance that appears (Lab).

Note that the **Device** field is set to `/dev/sdf`. You will use this device identifier in a later task.



Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
vol-0bcb3b55b22bf9c21 (My Volume)

Availability Zone
us-east-1a

Instance [Info](#)
i-0937e2f0035489ce9

Only instances in the same Availability Zone as the selected volume are displayed.

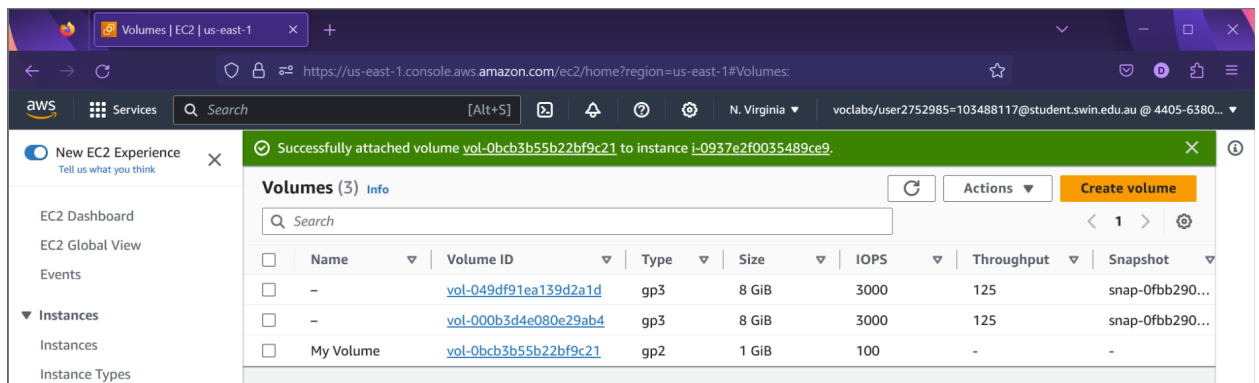
Device name [Info](#)
/dev/sdf

Recommended device names for Linux: /dev/sda1 for root volume, /dev/sd[f-p] for data volumes.

ⓘ Newer Linux kernels may rename your devices to `/dev/xvdf` through `/dev/xvdp` internally, even when the device name entered here (and shown in the details) is `/dev/sdf` through `/dev/sdp`.

Cancel **Attach volume**

14. Choose **Attach volume**. The volume state is now *In-use*.



Volumes (3) [Info](#)

Search

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot
☐	-	vol-049df91ea139d2a1d	gp3	8 GiB	3000	125	snap-0fbb290...
☐	-	vol-000b3d4e080e29ab4	gp3	8 GiB	3000	125	snap-0fbb290...
☐	My Volume	vol-0bcb3b55b22bf9c21	gp2	1 GiB	100	-	-

Task 3: Connect to Your Amazon EC2 Instance

Windows Users: Using SSH to Connect

These instructions are for Windows users only.

If you are using macOS or Linux, [skip to the next section](#).

15. Read through the three bullet points in this step before you start to complete the actions, because you will not be able see these instructions when the Details panel is open.
 - Choose the Details drop down menu above these instructions you are currently reading, and then choose Show. A Credentials window will open.
 - Choose the **Download PPK** button and save the **labsuser.ppk** file. Typically your browser will save it to the Downloads directory.
 - Then exit the Details panel by choosing the **X**.



16. Download needed software.

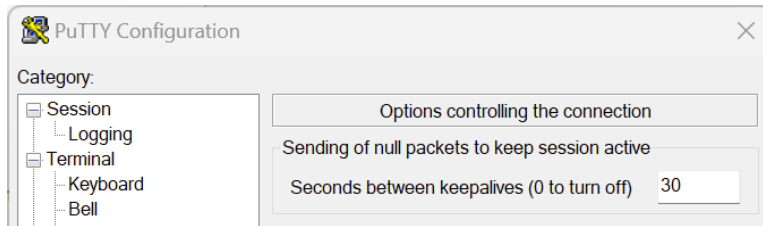
- You will use **PuTTY** to SSH to Amazon EC2 instances. If you do not have PuTTY installed on your computer, [download it here](#).



17. Open **putty.exe**

18. Configure PuTTY to not timeout:

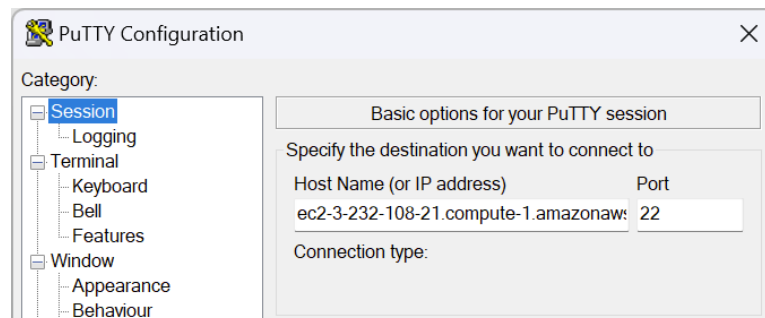
- Choose **Connection**
- Set **Seconds between keepalives** to **30**



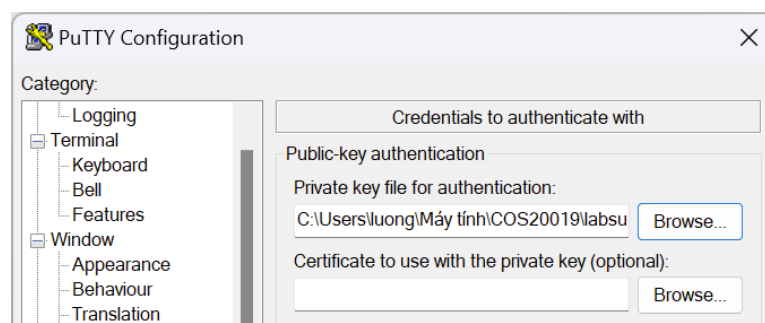
19. This allows you to keep the PuTTY session open for a longer period of time.

20. Configure your PuTTY session:

- Choose **Session**
- **Host Name (or IP address):** Paste the *Public DNS or IPv4 address* of the Lab instance that you noted earlier.



- Back in PuTTY, in the **Connection** list, expand **SSH**
- Choose **Auth** and expand **Credentials**
- Under **Private key file for authentication:** Choose **Browse**
- Browse to the *labsuser.ppk* file that you downloaded, select it, and choose **Open**
- Choose **Open** again



Name: Trac Duc Anh Luong - ID: 103488117

20. To trust and connect to the host, choose **Accept**.
21. When prompted **login as**, enter: **ec2-user**

This will connect you to the EC2 instance.

[illegible]

22. [Windows Users: Choose here to skip ahead to the next task.](#)

Name: Trac Duc Anh Luong - ID: 103488117

Create an ext3 file system on the new volume:

`sudo mkfs -t ext3 /dev/sdf`

```
ec2-user@ip-10-1-11-22:~  
/m/  
Last login: Fri Sep 29 23:31:56 2023 from 116.96.47.180  
[ec2-user@ip-10-1-11-22 ~]$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
devtmpfs        4.0M   0  4.0M   0% /dev  
tmpfs           475M   0  475M   0% /dev/shm  
tmpfs           190M  2.9M  188M   2% /run  
/dev/xvda1      8.0G  1.5G  6.5G  19% /  
tmpfs           475M   0  475M   0% /tmp  
/dev/xvda128    10M   1.3M  8.7M  13% /boot/efi  
tmpfs           95M   0   95M   0% /run/user/1000  
[ec2-user@ip-10-1-11-22 ~]$ sudo mkfs -t ext3 /dev/sdf  
mke2fs 1.46.5 (30-Dec-2021)  
Creating filesystem with 262144 4k blocks and 65536 inodes  
Filesystem UUID: 07f5840c-1fc3-4961-8932-64cb9flac97d  
Superblock backups stored on blocks:  
    32768, 98304, 163840, 229376  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (8192 blocks): done  
Writing superblocks and filesystem accounting information: done  
[ec2-user@ip-10-1-11-22 ~]$
```

Create a directory for mounting the new storage volume:

`sudo mkdir /mnt/data-store`

Mount the new volume:

`sudo mount /dev/sdf /mnt/data-store`

To configure the Linux instance to mount this volume whenever the instance is started, you will need to add a line to `/etc/fstab`.

`echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab`

View the configuration file to see the setting on the last line:

`cat /etc/fstab`

```
ec2-user@ip-10-1-11-22:~  
mke2fs 1.46.5 (30-Dec-2021)  
Creating filesystem with 262144 4k blocks and 65536 inodes  
Filesystem UUID: 07f5840c-1fc3-4961-8932-64cb9flac97d  
Superblock backups stored on blocks:  
    32768, 98304, 163840, 229376  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (8192 blocks): done  
Writing superblocks and filesystem accounting information: done  
  
[ec2-user@ip-10-1-11-22 ~]$ sudo mkdir /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$ sudo mount /dev/sdf /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$ echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$ cat /etc/fstab  
#  
UUID=9bbbebd1-cfa0-4f0c-b851-2dacbac7cf3c / xfs defaults,noatime 1 1  
UUID=68DE-EC82 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$
```

Name: Trac Duc Anh Luong - ID: 103488117

View the available storage again:

`df -h`

The output will now contain an additional line - `/dev/xvdf`:

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	484M	0	484M	0%	/dev
tmpfs	492M	0	492M	0%	/dev/shm
tmpfs	492M	460K	491M	1%	/run
tmpfs	492M	0	492M	0%	/sys/fs/cgroup
/dev/xvda1	8.0G	1.5G	6.6G	19%	/
tmpfs	99M	0	99M	0%	/run/user/0
tmpfs	99M	0	99M	0%	/run/user/1000
/dev/xvdf	976M	1.3M	924M	1%	/mnt/data-store

```
ec2-user@ip-10-1-11-22:~  
[ec2-user@ip-10-1-11-22 ~]$ sudo mkdir /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$ sudo mount /dev/sdf /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$ echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$ cat /etc/fstab  
#  
UUID=9bbbcbd1-cfa0-4f0c-b851-2dacbac7cf3c / xfs defaults,noatime 1 1  
UUID=68DE-EC82 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
devtmpfs        4.0M   0  4.0M   0% /dev  
tmpfs           475M   0  475M   0% /dev/shm  
tmpfs           190M  2.9M  188M   2% /run  
/dev/xvda1      8.0G  1.5G  6.5G  19% /  
tmpfs           475M   0  475M   0% /tmp  
/dev/xvda128    10M   1.3M   8.7M  13% /boot/efi  
tmpfs           95M    0    95M   0% /run/user/1000  
/dev/xvdf       975M  60K  924M   1% /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$
```

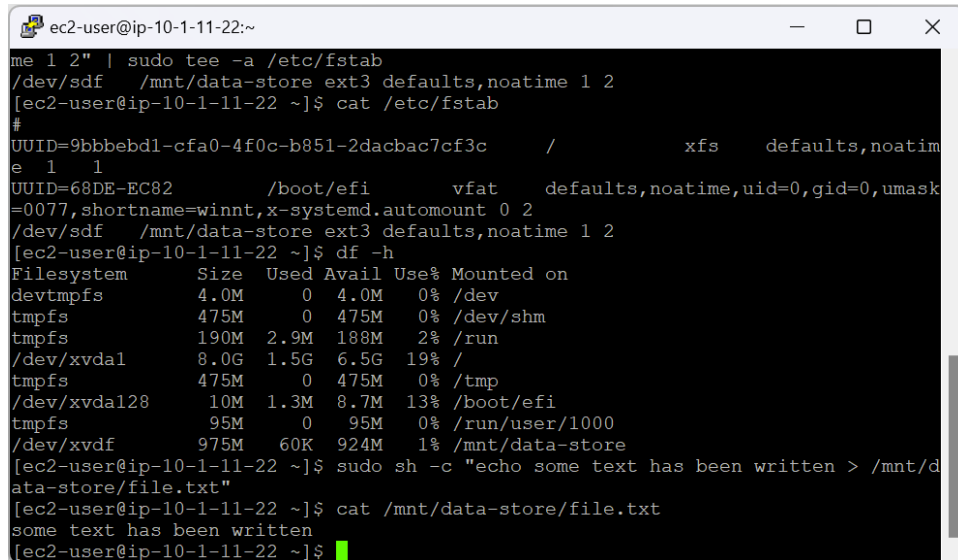
Name: Trac Duc Anh Luong - ID: 103488117

On your mounted volume, create a file and add some text to it.

`sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"`

Verify that the text has been written to your volume.

30. `cat /mnt/data-store/file.txt`



```
ec2-user@ip-10-1-11-22:~  
me 1 2" | sudo tee -a /etc/fstab  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$ cat /etc/fstab  
#  
UUID=9bbbebd1-cfa0-4f0c-b851-2dacbac7cf3c / xfs defaults,noatime  
e 1 1  
UUID=68DE-EC82 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask  
=0077,shortname=winnt,x-systemd.automount 0 2  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
devtmpfs        4.0M   0  4.0M   0% /dev  
tmpfs           475M   0  475M   0% /dev/shm  
tmpfs           190M  2.9M  188M   2% /run  
/dev/xvda1      8.0G  1.5G  6.5G  19% /  
tmpfs           475M   0  475M   0% /tmp  
/dev/xvda128    10M  1.3M  8.7M  13% /boot/efi  
tmpfs           95M   0   95M   0% /run/user/1000  
/dev/xvdf       975M  60K  924M   1% /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$ sudo sh -c "echo some text has been written > /mnt/d  
ata-store/file.txt"  
[ec2-user@ip-10-1-11-22 ~]$ cat /mnt/data-store/file.txt  
some text has been written  
[ec2-user@ip-10-1-11-22 ~]$
```

Task 5: Create an Amazon EBS Snapshot

In this task, you will create a snapshot of your EBS volume.

You can create any number of point-in-time, consistent snapshots from Amazon EBS volumes at any time. Amazon EBS snapshots are stored in Amazon S3 with high durability. New Amazon EBS volumes can be created out of snapshots for cloning or restoring backups. Amazon EBS snapshots can also be easily shared among AWS users or copied over AWS regions.

41. In the **AWS Management Console**, choose **Volumes** and select **My Volume**.

42. In the **Actions** menu, select **Create snapshot**.

43. Choose Add tag then configure:

- **Key:** Name
- **Value:** My Snapshot
- Choose Create snapshot

The screenshot shows the AWS Management Console interface for creating an EBS snapshot. The browser address bar indicates the URL: `https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSnapshotFromVolume:vol-0bcb3b55b22bf9c21`. The console breadcrumb navigation shows `EC2 > Volumes > vol-0bcb3b55b22bf9c21 > Create snapshot`. The main heading is **Create snapshot** with an **Info** icon. Below the heading is a sub-header: **Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.**

The **Details** section contains the following fields:

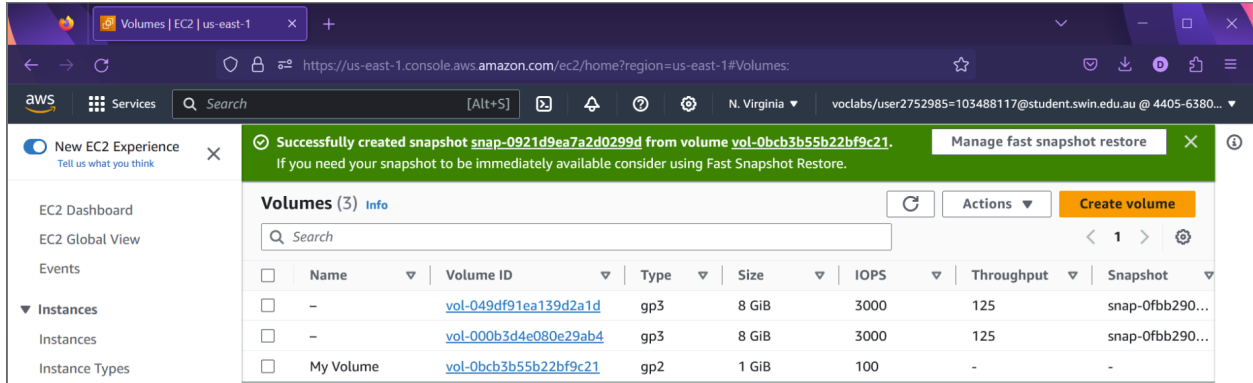
- Volume ID:** `vol-0bcb3b55b22bf9c21 (My Volume)`
- Description:** A text input field with the placeholder "Add a description for your snapshot" and a note "255 characters maximum."
- Encryption:** **Not encrypted** (with an **Info** icon)

The **Tags** section includes an **Info** icon and a description: "A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs." Below this is a table for adding tags:

Key	Value - optional	
Name	My Snapshot	Remove

Below the table is an **Add tag** button and a note: "You can add 49 more tags." At the bottom right of the form are **Cancel** and **Create snapshot** buttons.

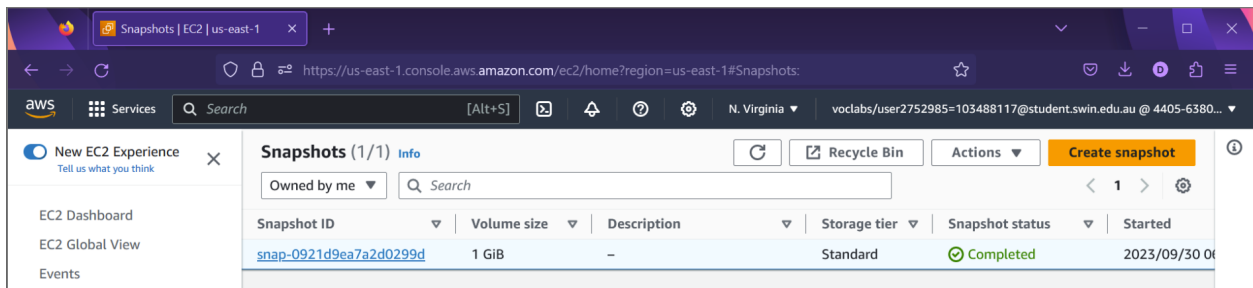
Name: Trac Duc Anh Luong - ID: 103488117



44. In the left navigation pane, choose **Snapshots**.

Your snapshot is displayed. The status will first have a state of *Pending*, which means that the snapshot is being created. It will then change to a state of *Completed*.

Note: Only used storage blocks are copied to snapshots, so empty blocks do not occupy any snapshot storage space.



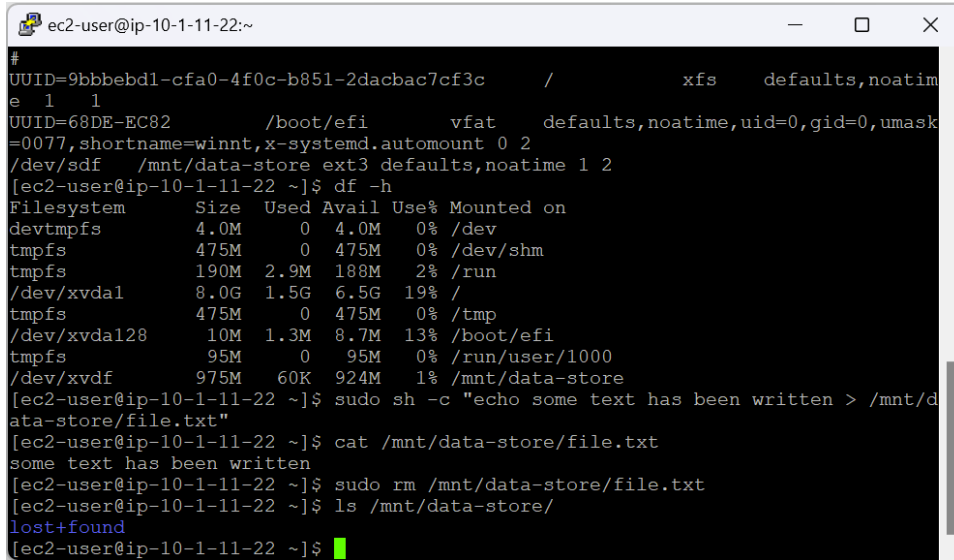
45. In your remote SSH session, delete the file that you created on your volume.

42. `sudo rm /mnt/data-store/file.txt`

43. Verify that the file has been deleted.

44. `ls /mnt/data-store/`

Name: Trac Duc Anh Luong - ID: 103488117

A terminal window titled 'ec2-user@ip-10-1-11-22:~' with standard window controls. It displays the output of the 'df -h' command, showing disk usage for various file systems. The user then runs 'sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"', followed by 'cat /mnt/data-store/file.txt' which outputs 'some text has been written'. Finally, the user runs 'sudo rm /mnt/data-store/file.txt' and 'ls /mnt/data-store/' which returns 'lost+found'.

```
#
UUID=9bbbebd1-cfa0-4f0c-b851-2dacbac7cf3c    /                xfs      defaults,noatim
e 1 1
UUID=68DE-EC82    /boot/efi        vfat      defaults,noatime,uid=0,gid=0,umask
=0077,shortname=winnt,x-systemd.automount 0 2
/dev/sdf    /mnt/data-store ext3 defaults,noatime 1 2
[ec2-user@ip-10-1-11-22 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           475M   0  475M   0% /dev/shm
tmpfs           190M  2.9M  188M   2% /run
/dev/xvda1      8.0G  1.5G  6.5G  19% /
tmpfs           475M   0  475M   0% /tmp
/dev/xvda128     10M  1.3M   8.7M  13% /boot/efi
tmpfs           95M   0   95M   0% /run/user/1000
/dev/xvdf       975M  60K  924M   1% /mnt/data-store
[ec2-user@ip-10-1-11-22 ~]$ sudo sh -c "echo some text has been written > /mnt/d
ata-store/file.txt"
[ec2-user@ip-10-1-11-22 ~]$ cat /mnt/data-store/file.txt
some text has been written
[ec2-user@ip-10-1-11-22 ~]$ sudo rm /mnt/data-store/file.txt
[ec2-user@ip-10-1-11-22 ~]$ ls /mnt/data-store/
lost+found
[ec2-user@ip-10-1-11-22 ~]$
```

Your file has been deleted.

Task 6: Restore the Amazon EBS Snapshot

If you ever wish to retrieve data stored in a snapshot, you can **Restore** the snapshot to a new EBS volume.

Create a Volume Using Your Snapshot

53. In the **AWS Management Console**, select **My Snapshot**.
54. In the **Actions** menu, select **Create volume from snapshot**.
55. For **Availability Zone** Select the same availability zone that you used earlier.
56. Choose Add tag then configure:
 - **Key:** **Name**
 - **Value:** **Restored Volume**
 - Choose Create volume

Name: Trac Duc Anh Luong - ID: 103488117

The screenshot shows the 'Create volume' page in the AWS Management Console. The breadcrumb navigation is 'EC2 > Snapshots > snap-0921d9ea7a2d0299d > Create volume'. The page title is 'Create volume' with an 'Info' link. Below the title is a sub-header: 'Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.'

The 'Volume settings' section contains the following fields:

- Snapshot ID:** snap-0921d9ea7a2d0299d (My Snapshot)
- Volume type:** General Purpose SSD (gp2)
- Size (GiB):** 1 (with a range from 1 to 16384 GiB)
- IOPS:** 100 / 3000 (Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.)
- Throughput (MiB/s):** Not applicable
- Availability Zone:** us-east-1a
- Fast snapshot restore:** Not enabled for selected snapshot
- Encryption:** ☐ Encrypt this volume

The 'Tags - optional' section includes a table for adding tags:

Key	Value - optional	
Name	Restored Volume	Remove

Buttons at the bottom include 'Cancel' and 'Create volume'.

The screenshot shows the 'Snapshots' page in the AWS Management Console. A green banner at the top states: 'Successfully created volume vol-00c26ed5a4f19158a.' Below this, the 'Snapshots (1)' section shows a table with one entry:

	Name	Snapshot ID	Volume size	Description	Storage tier	Snapshot status
☐	My Snapshot	snap-0921d9ea7a2d0299d	1 GiB	-	Standard	Complete

The page also includes a 'New EC2 Experience' sidebar on the left and a 'Create snapshot' button in the top right of the snapshots list.

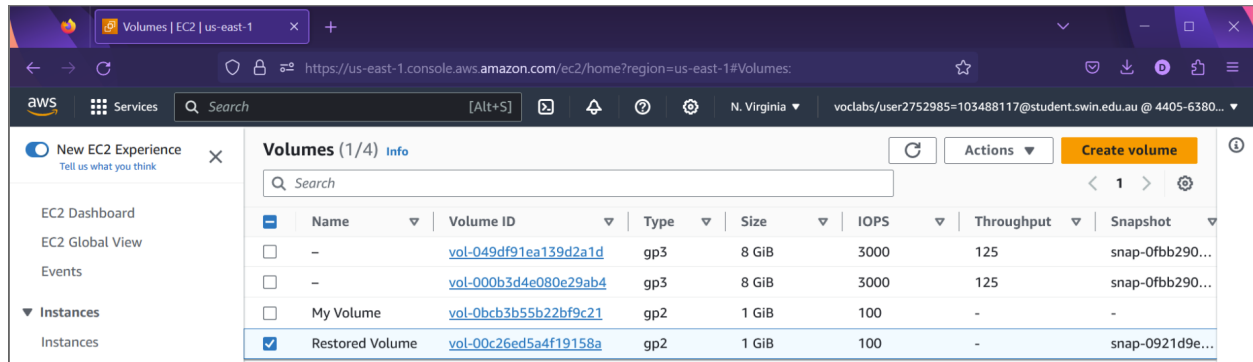
Name: Trac Duc Anh Luong - ID: 103488117

57. Note: When restoring a snapshot to a new volume, you can also modify the configuration, such as changing the volume type, size or Availability Zone.

Attach the Restored Volume to Your EC2 Instance

58. In the left navigation pane, choose **Volumes**.

59. Select **Restored Volume**.



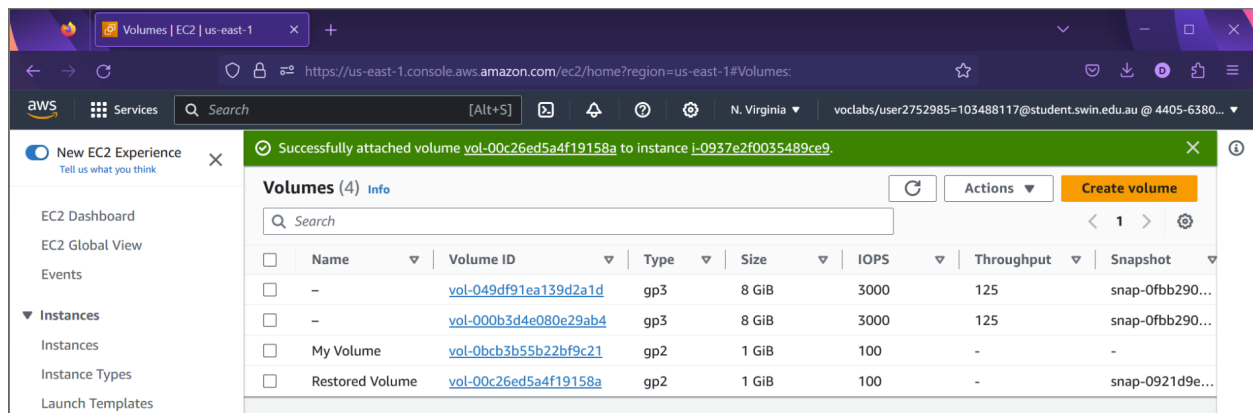
60. In the **Actions** menu, select **Attach volume**.

61. Choose the **Instance** field, then select the (Lab) instance that appears.

Note that the **Device** field is set to `/dev/sdg`. You will use this device identifier in a later task.

62. Choose **Attach volume**

The volume state is now *in-use*.



Mount the Restored Volume

63. Create a directory for mounting the new storage volume:

64. `sudo mkdir /mnt/data-store2`

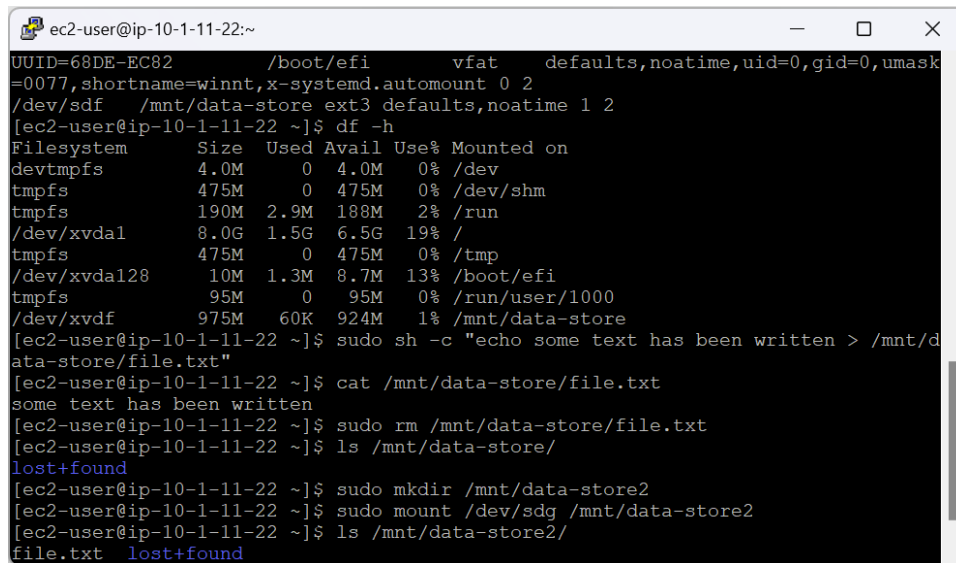
Name: Trac Duc Anh Luong - ID: 103488117

Mount the new volume:

```
sudo mount /dev/sdg /mnt/data-store2
```

Verify that volume you mounted has the file that you created earlier.

```
ls /mnt/data-store2/
```



```
ec2-user@ip-10-1-11-22:~  
UUID=68DE-EC82 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask  
=0077,shortname=winnt,x-systemd.automount 0 2  
/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2  
[ec2-user@ip-10-1-11-22 ~]$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
devtmpfs        4.0M   0  4.0M   0% /dev  
tmpfs           475M   0  475M   0% /dev/shm  
tmpfs           190M  2.9M  188M   2% /run  
/dev/xvda1      8.0G  1.5G  6.5G  19% /  
tmpfs           475M   0  475M   0% /tmp  
/dev/xvda128    10M  1.3M  8.7M  13% /boot/efi  
tmpfs           95M   0   95M   0% /run/user/1000  
/dev/xvdf       975M  60K  924M   1% /mnt/data-store  
[ec2-user@ip-10-1-11-22 ~]$ sudo sh -c "echo some text has been written > /mnt/d  
ata-store/file.txt"  
[ec2-user@ip-10-1-11-22 ~]$ cat /mnt/data-store/file.txt  
some text has been written  
[ec2-user@ip-10-1-11-22 ~]$ sudo rm /mnt/data-store/file.txt  
[ec2-user@ip-10-1-11-22 ~]$ ls /mnt/data-store/  
lost+found  
[ec2-user@ip-10-1-11-22 ~]$ sudo mkdir /mnt/data-store2  
[ec2-user@ip-10-1-11-22 ~]$ sudo mount /dev/sdg /mnt/data-store2  
[ec2-user@ip-10-1-11-22 ~]$ ls /mnt/data-store2/  
file.txt  lost+found
```

You should see file.txt.